

Lesson 15: Laplace Transforms - Solving Differential Equations

Professor Dave Explains Differential Equations Master Series · Lesson 15

Video: Laplace Transforms Part 1: Solving Differential Equations Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave describes the Laplace Transform as a magical time-machine for calculus. Differentiating and integrating functions in the time domain t is notoriously tedious. The Laplace transform acts as a mathematical lens that maps functions into the complex frequency domain s , where differentiation transforms into simple algebraic multiplication by s ! We solve the problem using basic high-school algebra, and then use inverse transform lookup tables to snap back into the time domain.

2. Core Theoretical Framework & Rigorous Formulations

1. Definition & Elementary Transforms

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt \quad (s > 0)$$

Time Domain $f(t)$	Laplace Domain $F(s)$
1	$\frac{1}{s}$
$t^n \ (n \in \mathbb{Z}^+)$	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(kt)$	$\frac{k}{s^2+k^2}$
$\cos(kt)$	$\frac{s}{s^2+k^2}$
$e^{at}f(t)$ (First Shift Theorem)	$F(s-a)$

2. Transform of Derivatives

$$\mathcal{L}\{y'\} = sY(s) - y(0), \quad \mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$$

3. Fully Solved Whiteboard Problems

Concordia Final Exam Problem: Solving a Second-Order IVP via Laplace

Problem Statement: Solve $y'' - 4y' + 4y = 6e^{2t}$ with $y(0) = 1, y'(0) = 0$ using Laplace transforms.

Step 1: Apply Laplace Transform to Both Sides

$$\mathcal{L}\{y''\} - 4\mathcal{L}\{y'\} + 4\mathcal{L}\{y\} = 6\mathcal{L}\{e^{2t}\}. [s^2Y(s) - s(1) - 0] - 4[sY(s) - 1] + 4Y(s) = \frac{6}{s-2}.$$

Step 2: Collect Algebraic Terms in $Y(s)$

$$(s^2 - 4s + 4)Y(s) - s + 4 = \frac{6}{s-2} \implies (s-2)^2Y(s) = s - 4 + \frac{6}{s-2}.$$

Step 3: Combine and Isolate $Y(s)$

$$(s-2)^2Y(s) = \frac{(s-4)(s-2)+6}{s-2} = \frac{s^2-6s+8+6}{s-2} = \frac{s^2-6s+14}{s-2}. \text{ Therefore: } Y(s) = \frac{s^2-6s+14}{(s-2)^3}.$$

Step 4: Shift Algebra to $u = s - 2$

$$\text{Let } s = (s-2) + 2: s^2 - 6s + 14 = [(s-2) + 2]^2 - 6[(s-2) + 2] + 14 = (s-2)^2 + 4(s-2) + 4 - 6(s-2) - 12 + 14 = (s-2)^2 - 2(s-2) + 6. \text{ Therefore: } Y(s) = \frac{(s-2)^2 - 2(s-2) + 6}{(s-2)^3} = \frac{1}{s-2} - \frac{2}{(s-2)^2} + \frac{6}{(s-2)^3}.$$

Step 5: Apply Inverse Laplace Transform with First Shift Theorem

$$\text{Recall } \mathcal{L}^{-1}\{\frac{1}{s^3}\} = \frac{t^2}{2}. \text{ By the shift theorem: } y(t) = e^{2t} - 2te^{2t} + 6\left(\frac{t^2}{2}\right)e^{2t} = e^{2t}(1 - 2t + 3t^2).$$

Common Exam Pitfall & Trapdoor Avoidance

Always include initial conditions when transforming derivatives: $\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0)$. Forgetting the minus sign on $-[-y(0)]$ causes a sign inversion error in $Y(s)$.

4. Key Takeaways & Exam Strategy Formula Card

Core Takeaway & Exam Rule

Laplace transforms bake initial conditions directly into the algebraic setup, bypassing undetermined coefficients entirely.

- Video reference: [Laplace Transforms Part 1: Solving Differential Equations](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.